



Advanced Linear Algebra: Foundations to Frontiers

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2.5.1 Additional homework ¶

Homework 2.5.1.1. $U \in \mathbb{C}^{m \times m}$ is unitary if and only if $(Ux)^H(Uy) = x^H y$ for all $x, y \in \mathbb{C}^m$.

▼ Hint

Revisit the proof of Homework 2.2.4.6.

Homework 2.5.1.2. Let $A, B \in \mathbb{C}^{m \times n}$. Furthermore, let $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ be unitary.

TRUE/FALSE: $UAV^H = B$ iff $U^H BV = A$.

▼ Answer

TRUE

Now prove it!

Homework 2.5.1.3. Prove that nonsingular $A \in \mathbb{C}^{n \times n}$ has condition number $\kappa_2(A) = 1$ if and only if $A = \sigma Q$ where Q is unitary and $\sigma \in \mathbb{R}$ is positive.

▼ Hint

Use the SVD of A .

Homework 2.5.1.4. Let $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ be unitary.

ALWAYS/SOMETIMES/NEVER: The matrix $\left(\begin{array}{c|c} U & 0 \\ \hline 0 & V \end{array} \right)$ is unitary.

▼ Answer

ALWAYS

Now prove it!

Homework 2.5.1.5. Matrix $A \in \mathbb{R}^{m \times m}$ is a stochastic matrix if and only if it is nonnegative (all its entries are nonnegative) and the entries in its columns sum to one: $\sum_{0 \leq i < m} \alpha_{i,j} = 1$. Such matrices are at the core of Markov processes. Show that a matrix A is both unitary matrix and a stochastic matrix if and only if it is a permutation matrix.

Homework 2.5.1.6. Show that if $\|\cdot\|$ is a norm and A is nonsingular, then $\|\cdot\|_{A^{-1}}$ defined by $\|x\|_{A^{-1}} = \|A^{-1}x\|$ is a norm.

Interpret this result in terms of the change of basis of a vector.

Homework 2.5.1.7. Let $A \in \mathbb{C}^{m \times m}$ be nonsingular and $A = U\Sigma V^H$ be its SVD with

$$\Sigma = \begin{pmatrix} \sigma_0 & 0 & \cdots & 0 \\ 0 & \sigma_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{m-1} \end{pmatrix}$$

The condition number of A is given by (mark all correct answers):

1. $\kappa_2(A) = \|A\|_2 \|A^{-1}\|_2$.
2. $\kappa_2(A) = \sigma_0 / \sigma_{m-1}$.
3. $\kappa_2(A) = u_0^H A v_0 / u_{m-1}^H A v_{m-1}$.
4. $\kappa_2(A) = \max_{\|x\|_2=1} \|Ax\|_2 / \min_{\|x\|_2=1} \|Ax\|_2$.

(Mark all correct answers.)

Homework 2.5.1.8. [Theorem 2.2.4.4](#) stated:

If $A \in \mathbb{C}^{m \times m}$ preserves length ($\|Ax\|_2 = \|x\|_2$ for all $x \in \mathbb{C}^m$), then A is unitary. Give an alternative proof using the SVD.

Homework 2.5.1.9. In [Homework 1.3.7.2](#) you were asked to prove that $\|A\|_2 \leq \|A\|_F$ given $A \in \mathbb{C}^{m \times n}$. Give an alternative proof that leverages the SVD.

Homework 2.5.1.10. In [Homework 1.3.7.3](#), we skipped how the 2-norm bounds the Frobenius norm. We now have the tools to do so elegantly: Prove that, given $A \in \mathbb{C}^{m \times n}$,

$$\|A\|_F \leq \sqrt{r} \|A\|_2,$$

where r is the rank of matrix A .